

Bridging Connectomics and Computation: Analyzing the Dynamics of WSBM-Constrained Echo State Networks using InputDSA

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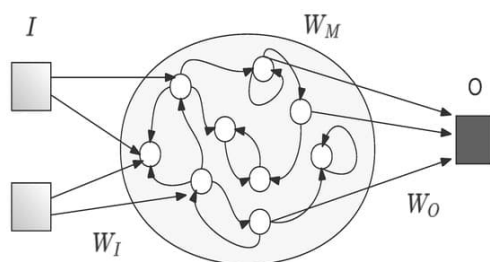
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Table of contents

1. Motivation
2. Methodology
3. Results
4. Conclusion

1. Motivation

Echo State Network (ESN)



Connectome-Constrained Model

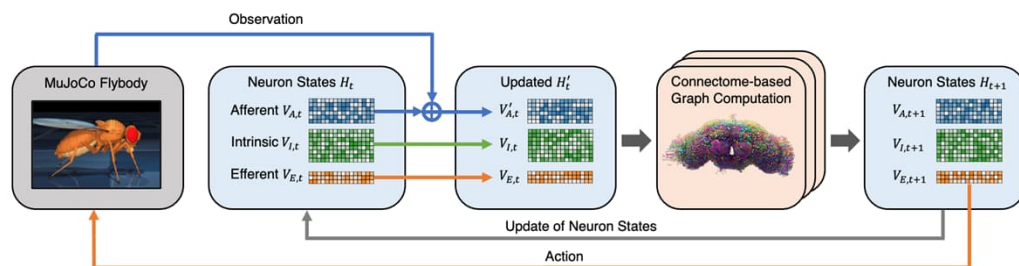


Figure from Jin et al - 2026 - Whole-Brain Connectomic Graph Model Enables Whole-Body Locomotion Control in Fruit Fly

Weighted Stochastic Block Model

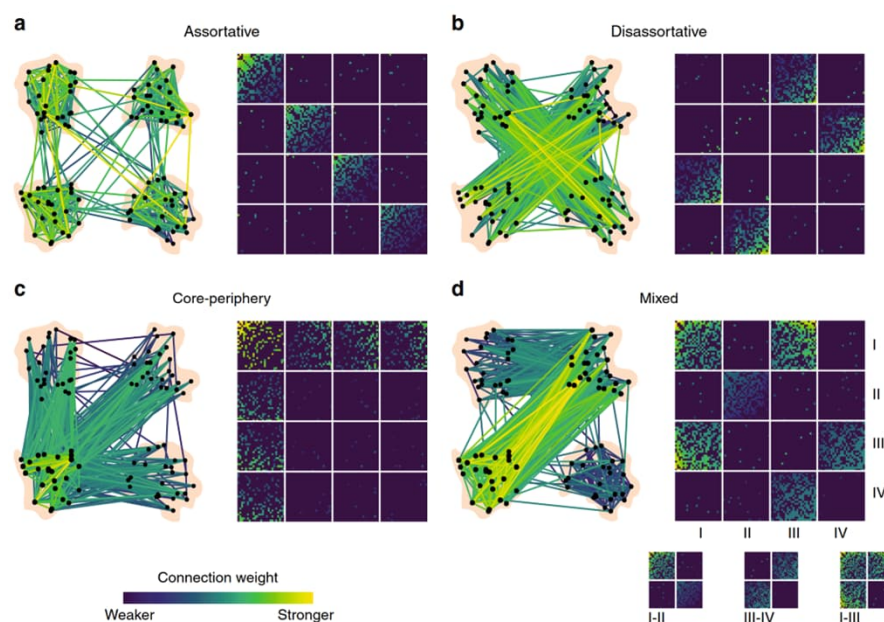


Figure from Betzel et al. - 2018 - Diversity of meso-scale architecture in human and non-human connectomes

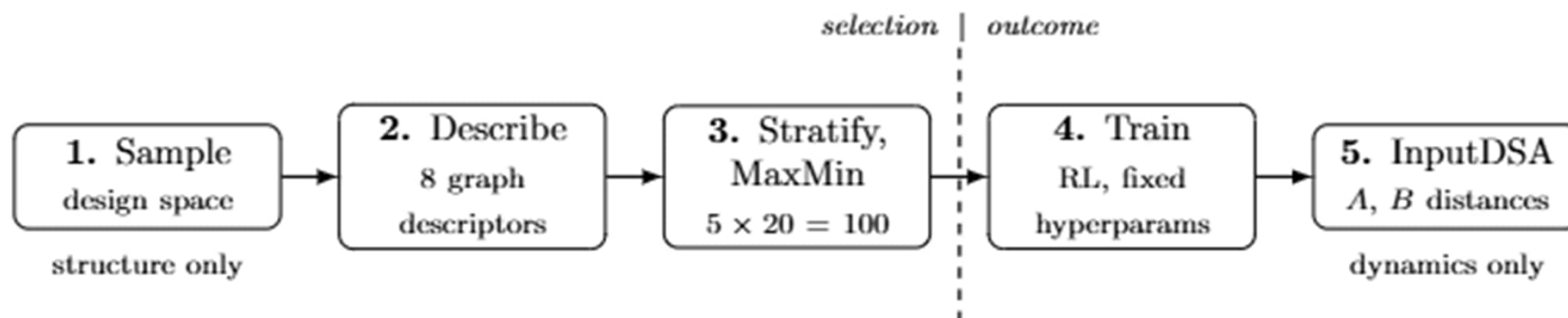
2. Methodology

$$W_{ij} \sim \underbrace{\mathcal{N}(\mu_{z_i, z_j}, \sigma_{z_i, z_j}^2)}_{\text{Connection's strength}} \cdot \underbrace{\text{Bernoulli}(\theta_{z_i, z_j})}_{\text{Connection's probability}}, \quad W_{ii} = 0,$$

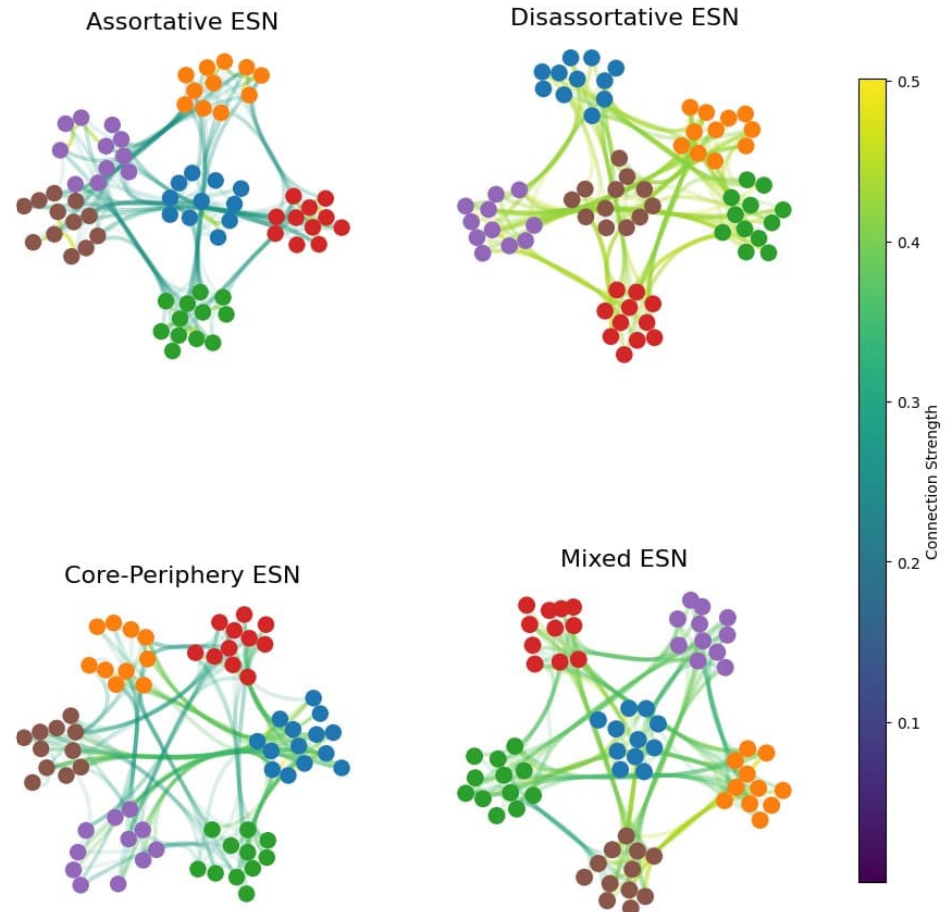
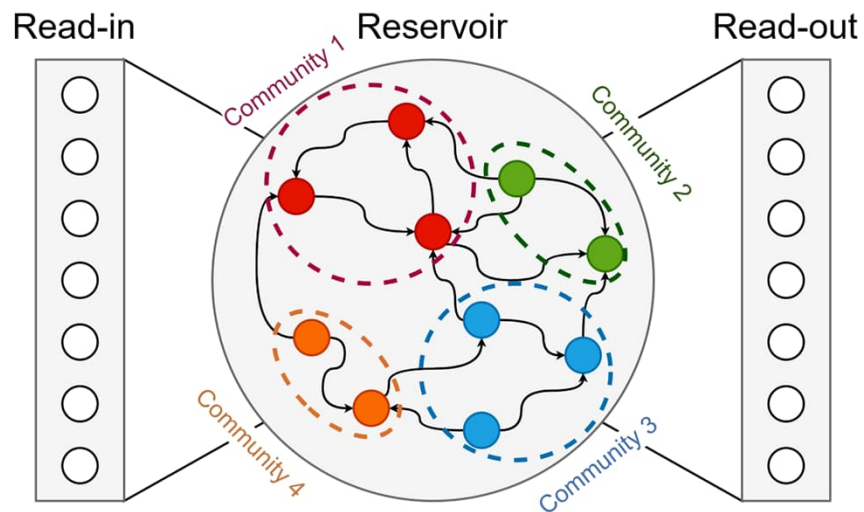
With:

- z_i and z_j are two different nodes in the reservoir
- μ, σ being mean and std associated with (z_i, z_j)
- θ is the chance to form a connection between z_i and z_j

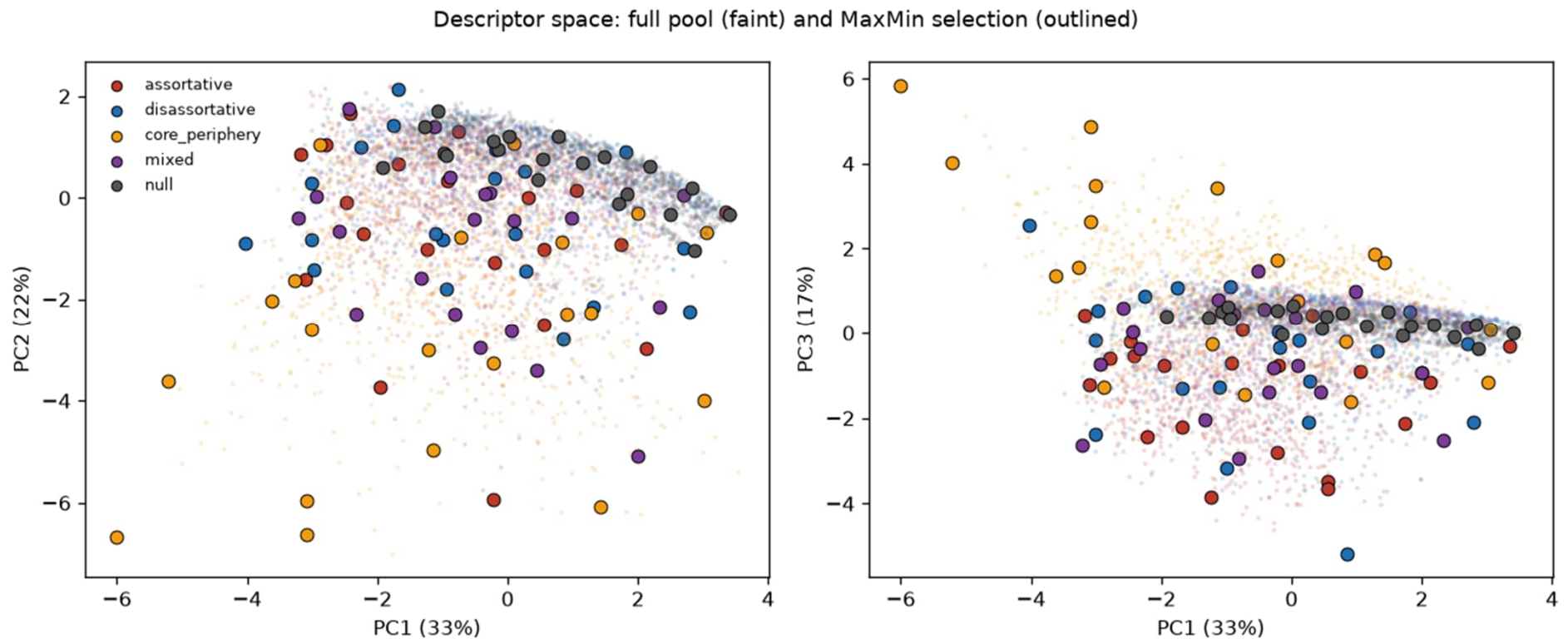
2. Methodology



2. Methodology



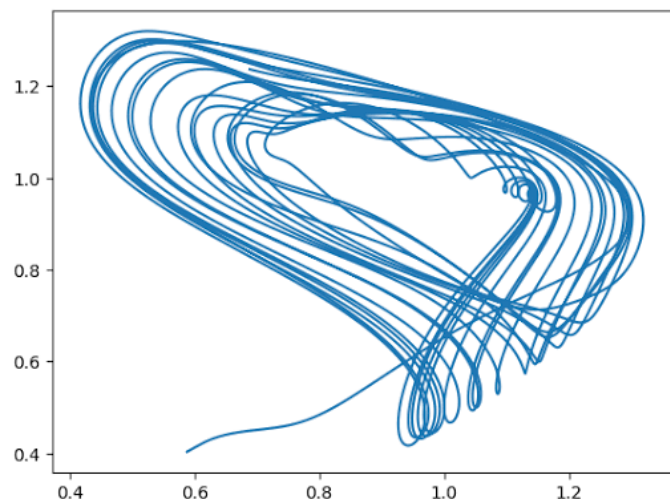
3. Results: Hyperparameter space



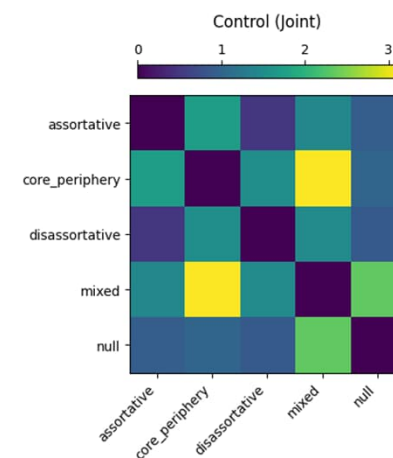
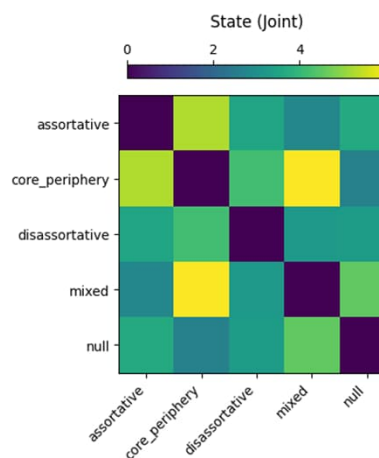
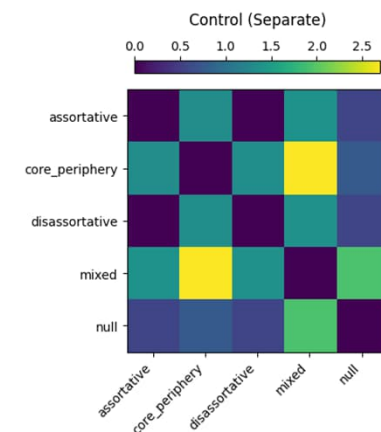
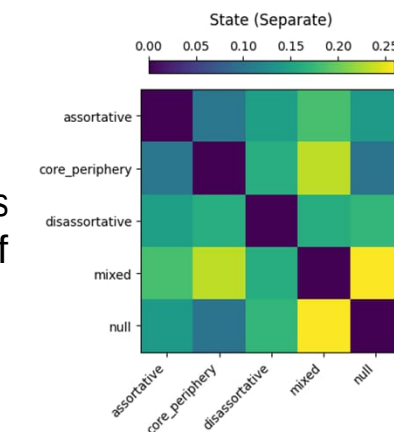
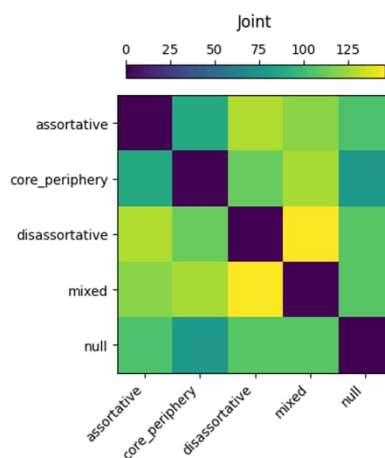
3. Results: InputDSA

Applied dataset: *Mackey-glass equation*

$$\frac{dy(t)}{dt} = 0.2 \frac{y(t-\tau)}{1 + y^{10}(t-\tau)} - 0.1y(t), y(0) = 1.5$$

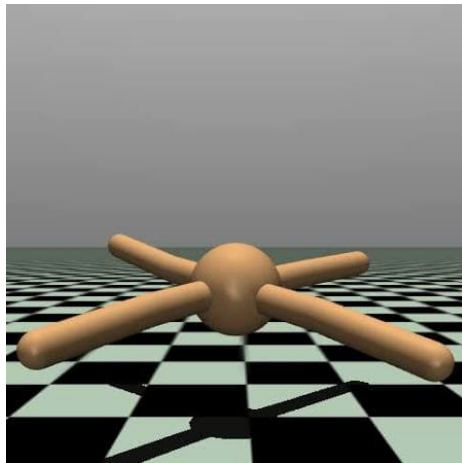


Dissimilarity matrices
of input and state of
5 systems

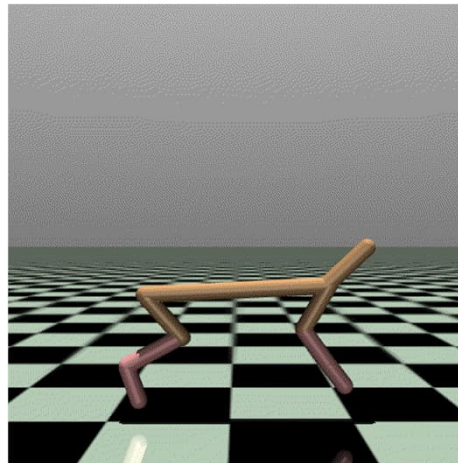


4. Conclusion

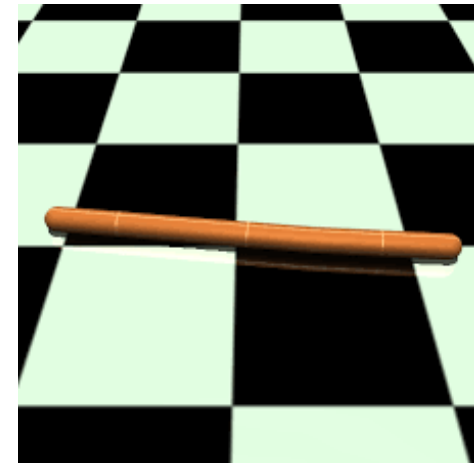
- Reinforcement Learning is the next step



Ant-v4



Cheetah-v4



Swimmer-v4

- Reservoirpy's ecosystem fully integration + PR



THANK YOU FOR LISTENING

Any questions?